EDA Report – Iris Dataset

Internship Project Submission

Prepared using Python, Pandas, Matplotlib, and Scikit-learn.

1. Objective

To perform Exploratory Data Analysis (EDA) on the Iris dataset and build a regression model to predict Sepal Length.

2. Dataset Overview

Rows: 150

Columns: 5

- SepalLengthCm
- SepalWidthCm
- PetalLengthCm
- PetalWidthCm
- Species

3. Data Cleaning & Preprocessing

Removed the Id column.

Total missing values: 0

Species column encoded for machine learning model.

4. Statistical Summary

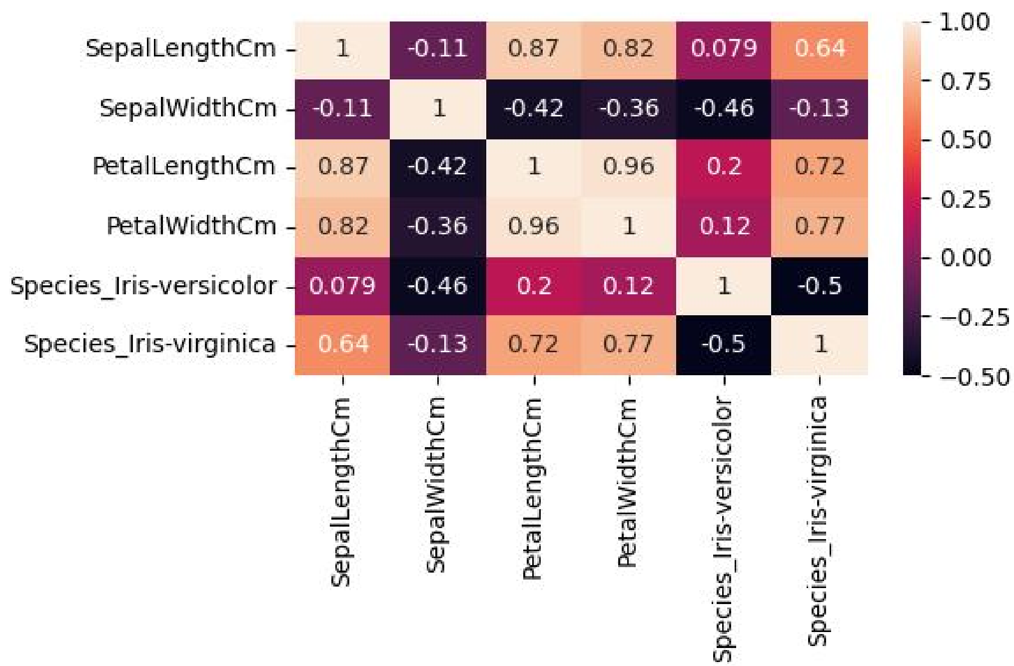
Metric	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.0	150.0	150.0	150.0
mean	5.84	3.05	3.76	1.2
std	0.83	0.43	1.76	0.76
min	4.3	2.0	1.0	0.1
25%	5.1	2.8	1.6	0.3
50%	5.8	3.0	4.35	1.3
75%	6.4	3.3	5.1	1.8
max	7.9	4.4	6.9	2.5

5. Key Findings

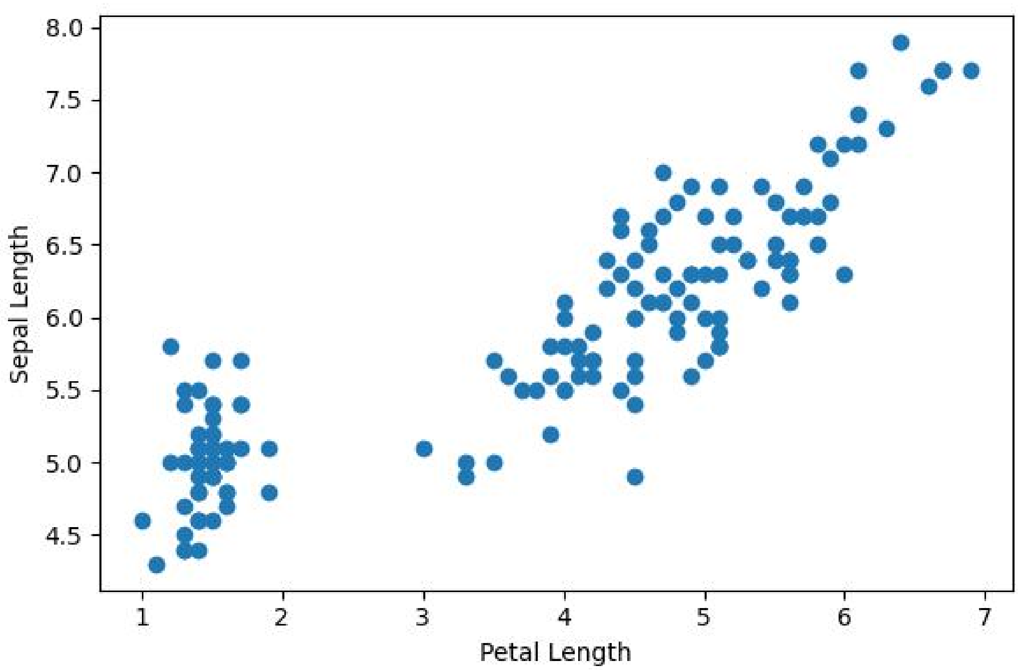
- No missing values were present in the dataset.
- Petal features show strong relationship with Sepal Length.
- Dataset is balanced across all flower species.
- Feature relationships are visually clear from correlation analysis.

6. Visualizations

Correlation Heatmap



Scatter Plot: Petal Length vs Sepal Length



7. Model Evaluation

- Mean Absolute Error (MAE): 0.228
- Root Mean Squared Error (RMSE): 0.304
- R^2 Score: 0.866

8. Conclusion

The Iris dataset was successfully analyzed using EDA techniques. The Linear Regression model achieved good prediction performance for Sepal Length based on flower measurements.